Core Concepts in Quantum Field Theory

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This note is a summary of formal core concepts in quantum field theory. Including Symmetry, Renormalization, Anomalies, Solitons and Instantons. The note is mainly based on Peskin and Schroeder's "An Introduction to Quantum Field Theory"[1], Tong's lecture notes on gauge theory, and some other references.

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Contents

CHAPTER 1

Fundamental Ideas

This chapter serves as an introduction to basic ideas in QFT, which will be intensively used and discussed in the following chapters. It will not be self contained, yet all preliminaries(free field, path integral, Feymann diagrams) will be found in basic QFT textbooks.

1 LSZ Reduction Formula

For a scattering theory, it is important predict the S-matrix as observable quantity. For QFT, LSZ reduction formula is the one that tells us how to **calculate the S-matrix from the correlation functions**. Then we can:

1. Use Feymann diagrams to calculate correlation functions (Both functional derivative method or canonical quantization with interaction picture).
2. Use LSZ reduction formula to get the S-matrix from the correlation functions.

1.1 Kallen-Lehmann Form of Propagator

This is a topic to ask:

- What will be the propagator consistent lorentz invariant quantum field theory?

We then can see the form of the propagator is highly constraint by lorentz invariant and the quantum mechanics structure, which turns out to be in a controlable form, called Kallen-Lehmann form.

1.1.1 Kallen-Lehmann Form of Scalar Propagator

1.1.2 Kallen-Lehmann Form of Fermion Propagator

1.1.3 Example: QED Fermion Propagator

Now we have a look at a concrete example of a lorentz invariant QFT, which is QED.

1.2 LSZ Reduction Formula

1.2.1 Derivation of LSZ Reduction Formula

1.2.2 S-Matrix from Feymann Diagrams

2 Ward-Takahashi Identity

Symmetry plans the fundamental role in QFT, and the Ward-Takahashi identity is a consequence of the symmetry. I'll first introducs the formal structure of the Ward-Takahashi identity following section 9.6 of [1], and then talk about the application in electrodynamics, following section 7.4 of [1].

3 Quantization of Gauge Fields

CHAPTER 2

Renormalization

Renormalization is the procedure to deal with divergences in quantum field theory. It is a systematic way to cancel the divergences and get finite results for physical quantities.

1 Regualization

Before starting with the systematic renormalization canceling out divergence, we need to first regularize the divergences, which is the process to make the divergences well defined. This is done by introducing a regularization scheme, which is a mathematical tool to make the divergences well defined.

1.1 Regularization Schemes

To regularize the divergences, we need to first calculate the loop integrals to some good extent, and the introduce the regularization scheme to make the divergences well defined. There are several regularization schemes, and the most common ones are:

- Pauli-Villars Regularization
- Dimensional Regularization

This section we will intensively study how to calculate loop integrals and do regularization in a standard way. Here are the general steps to folow:

1. Use **Feymann Parameters** to combine the denominators of the propagators in the loop integrals, which will make the loop integrals easier to calculate.
2. Make l

1.1.1 Feymann Parameterization

1.1.2 Dimensional Regularization

1.2 Examples

Lets use the mathematical tools developed in the previous sections to calculate some examples of loop integrals and do regularization, which will be useful for the following sections on renormalization.

1.2.1 Example: φ^4 Theory

1.2.2 Example: Electric Vertex Function

This is the most basic example of renormalization, which is the one-loop correction to the electric vertex function, given in section 6.3 and 7.2, 7.3 of [1].

1.2.3 Example: Vaccum Polarization

This is another example from Peskin and Schroeder [1], which is the one-loop correction to the photon propagator, given in section 7.5.

2 Renormalization

We have seen that the observables we care about seems to have divergences due to the loop integrals. Now we need to find a way to understand why there are these divergences and how to get finite results for physical quantities. This is the process of renormalization.

2.1 Bare Perturbation Theory

Before going to a modern and systematic way to do renormalization, we need to first try to do it in a more naive and cumbersome way called **bare perturbation theory**. Yet it may help us to understand why the systematica **renormalized perturbation theory** works.

2.1.1 General Method

Before we have encountered divergences in observables, seems like the theory is ill defined. However, in fact its is our understanding of the theory is incomplete, here is a modern understanding:

- Before, we have infinity due to the fact that we are using the unobservable infinite quantities $m_0, \lambda_0, \Lambda, Z, \ldots$ to express the S Matrix, thus the coefficients turns out to be infinite.
- We have to define some physical quantities (and argue why they are physical) and think of them as the result of stuffs we really measure in experiments, such as the physical mass m, the physical coupling constant λ …
- Now we can express the S Matrix in terms of the physical quantities, turns out that the coefficients are finite, and we can get finite results for physical quantities.

To do this, we follow the following steps:

1. **Regularize the divergences in Observable** (eg. S Matrix), and express the Observable in terms of the unobservable infinite quantities $m_0, \lambda_0, \Lambda, Z, \ldots$, with infinite divergences.
2. **Define the physical quantities**, such as the physical mass m, the physical coupling constant λ and calculate them perturbatively in terms of the unobservable infinite quantities $m_0, \lambda_0, \Lambda, Z, \ldots$ to some orders.
3. **Express the Observable in terms of the physical quantities**, and we will find that the coefficients are finite, and we can get finite results for physical quantities.

> **Note**
>
> It is important to have all steps in the same order of perturbation. For tree level, there are no loops and divergence, thus we can directly express bare quantity as physical quantities, if we fix the order of perturbation to be tree level.

2.1.2 Example: Electric Vertex Function

2.1.2.1 Divergence in all orders

There is a problem of can field strength renormalization cancel divergence to all orders? this is enforced by the Ward-Takahashi identity, which is a consequence of the gauge symmetry.

2.1.3 Example: Vaccum Polarization

2.2 Classifying Divergences

2.3 Renormalized Perturbation Theory

We have seen that the bare renormalization is cumbersome and may have some potential problems unclear in the procedure.

2.3.1 General Method

Renormalization Condition is in fact defining what the physical quantities are in this theory.

2.3.2 Understanding from LSZ

This subsection address understanding of renormalization from the LSZ reduction formula

2.4 Examples of RPT

2.4.1 Example: φ^4 Theory

CHAPTER 3

Renormalization Group

Before we deal with

CHAPTER 4

Anomalies

1 Chiral Anomalies

This section I will follow David Tong's lecture on gauge theory to talk about chiral anomalies in fermions when coupled to gauge fields.

2 Gauge Anomalies

3 ‘t Hooft Anomaly

CHAPTER 5

Symmetry Breaking

Bibliography

[1] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory*. Reading, Mass: Addison-Wesley Pub. Co, 1995.